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Usage of the optimal fuzzy system models on chloride diffusion of concrete

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Abstract

Interaction to chloride ions is only one of many challenging scenarios that reinforced concrete buildings may face during their lifespan. This exposure may cause concrete structures, particularly those near the seaside, to deteriorate and become less long-lasting. The chloride diffusion coefficient (*CD*) of nonsteady mode evident concrete may be accurately predicted using AI models that use empirical field data over an extended period of time. By isolating the most important factors, this technique has the potential to improve the evaluation of a concrete structure's durability. This work shows how to estimate the *CD* of concrete under different exposure situations using an adaptive neuro-fuzzy inference system (*ANFIS*). The Bald Eagle Search (*BES*), Black Widow Optimization (*BWO*), and artificial rabbit optimization (*ARO*) techniques were used to enhance the forecast models, which were trained on a dataset involving 216 data points. Findings indicate that the *ANFAR*, *ANFBW*, and *ANFBE* models have promising futures as *DC* predictors for concrete under different exposure conditions, all while maintaining suitable R^2 rates. The findings imply that *ANFAR*, *ANFBW*, and *ANFBE* may accurately anticipate concrete *CD* under diverse exposure scenarios. *ANFBE* attained R^2 values of 0.9959 throughout training and 0.997 throughout testing, larger than other systems in spite of their higher capability during prediction accuracy. When exposure time (*ET*) and exposure condition (*EC*) are eliminated from the input set, *RRSE*, *NMSE*, and R^2 values significantly rise and drop. Importantly, eliminating *ET* led to a remarkable impact on the target, where R^2 reduced from 0.997 to 0.7761 and *NMSE* increased from 0.001 to 0.1834 in testing.

Keywords: Durability, Chloride diffusion, Concrete, *ANFIS*, Optimization algorithms

Introduction

Because reinforced concrete (*RC*) is thought to be both affordable and long-lasting, it is often utilized in infrastructure. Nevertheless, *RC* structures—especially those found in coastal, marine, and offshore environments—face severe deterioration and maintenance expenses as a result of the chloride-induced problem. Reinforcement corrosion results from the reinforcement becoming destructive once the chloride has seeped into the concrete and achieved a threshold concentration around the reinforcing steel [1]. According to the performance method, one of the three endurance indicators for constructing reinforced concrete subjected to a maritime environment is the chloride diffusion coefficient [2]. It is challenging to empirically determine the diffusion coefficient or chloride

penetration in concrete for every project due to the time and cost involved. The chloride diffusion coefficient is necessary to construct all of the models for chloride intrusion into concrete forecasting models, including the multispecies ionic transport model [3, 4] and the geochemical model [5, 6]. Thus, it is critical to determine the concrete's chloride diffusion coefficient in order to improve the strength and longevity of buildings and to develop effective rehabilitation strategies. Multiple experimental models have been established to determine the chloride diffusion coefficient of concrete, such as ordinary concrete and concrete utilizing supplementary cementitious materials (*SCM*s) like fly ash (*FA*), ground granulated blast furnace slag (*GGBFS*), and silica fume (*SF*). This is done to minimize the cost and time-consuming nature of experimentation [7]. Nevertheless, because of the intricacy of the diffusion events in concrete, the experimental determination of the chloride diffusion coefficient utilizing *SCM*s is more difficult [8]. Consequently, it is challenging to establish an exact formula for forecasting the chloride diffusion coefficient.

The rapid advancement of artificial intelligence technology over the last numerous years has led to the widespread use of machine learning (*ML*) models in many facets of life [9]. *ML* techniques have shown to be helpful in solving a number of challenging civil engineering problems, including structural engineering [10], geotechnical engineering [11], and materials science [12]. To assess the chloride diffusion coefficient, some studies on concrete have employed *AI* techniques like artificial neural network (*ANN*) models and multigene genetic [13, 14]. For the purpose of predicting the chloride diffusion coefficient of concrete, the majority of research created *ANN* models. With the exception of Liu et al.'s study [14], the other studies' predictions for the chloride diffusion coefficient of regular concrete, high-performance concrete, and self-consolidating concrete ranged from 72 to 270 data points. In the Lui et al.'s study [14], inquiry 653 samples were utilized to construct *ANN* models for forecasting of the chloride diffusion coefficients in concrete containing various *SCM*s, including *GGBFS*, *FA*, and *SF*. The experiments' *ML* models performed very well in terms of precisely predicting the coefficient of chloride diffusion.

The research used two innovative types of ensemble *AI* methods—the genetic programming forest (*GPF*) and linear genetic programming forest (*LGPF*) techniques—to replicate the *CD* of concrete. Field data was collected for the research. As the control ensemble methodology, the random forest (*RF*) technique was used for comparison. The highest-performing *LGPF* model outperforms even the most sophisticated *GPF* and *RF* models. Furthermore, the findings show that the silica particle-to-binder ratio, the length of exposure, and the environmental factors all have a major impact on the durability coefficient of concrete [15]. The *ANN* developed four creature-dependent metaheuristic optimization techniques: the jellyfish search optimizer (*JSO*), the whale optimization algorithm (*WOA*), and the marine predators algorithm (*MPA*). The offered methods were used to mimic concrete's D_C under atmospheric, tidal, impact, and submerged circumstances. The research included 216 field trial data points. The synthesized approaches with simpler designs outperform the old approach. The Wilcoxon rank-sum test shows that *ANN-JSO* outperforms other *ANN* methods. Additionally, the mean *ANN-MPA*, *ANN-SSA*, and *ANN-WOA* examination results stay unaltered [16]. Research offered several *ML* techniques to estimate the chloride diffusion coefficient of concrete with *SCM*s like silica fume, ground granulated

blast furnace slag, and fly ash. Eight *ML* models—support vector machine (*SVM*), extreme learning machine (*ELM*), K-nearest neighbors (*KNN*), Light Gradient Boosting (*LGB*), eXtreme Gradient Boosting (*XGB*), random forest (*RF*), gradient boosting (*GB*), and AdaBoost (*AdB*)—were assessed after a database containing nine input parameters was created. The model that performed best in predicting the chloride diffusion coefficient was the gradient boosting model. The choice of the finest *ML* method gradient boosting helped construct a reliable soft computing tool for concrete structure durability design, including mix design optimization and binder selection [17]. Research illustrated the use of *ANFIS* for predicting the direct current of concrete. The predictive models were refined by the use of the Harris Hawks Optimization (*HHO*) and Chimp Optimization Algorithm (*COA*). The findings indicated that the *ANF* (*COA*) and *ANF* (*HHO*) models had significant potential for precisely forecasting the durability characteristics of concrete under diverse exposure conditions while maintaining suitable coefficients of determination (R^2). The models, by delivering accurate *CD* forecasts, might enhance maintenance methods, improve material selection, and facilitate the construction of more resilient concrete buildings, hence enhancing infrastructure durability and decreasing long-term maintenance expenses [18].

Because predicting the *CD* of concrete is a complex process with time-varying behavior, the traditional models based on short-term experimental study have many drawbacks. A few criticisms of the rapid chloride migration test consist of that it measures flow before achieving a steady migration state, raises the temperature of the sample because of high voltage, and measures flow for all ions, not only chloride ions. It takes a lot of time and money to conduct long-term field testing on the *CD* of concrete. This subject highlights the importance of the results from long-term, in-depth field research on the *CD* of concrete conducted over a number of years. We may get a better knowledge of the *CD* of concrete in practical situations and provide more reliable findings by combining the findings of this great research and using artificial intelligence techniques to analyze the gathered data as a complete database. Using data from previous studies, this study assembled a comprehensive database of long-term studies on the *CD* of concrete in marine settings. The *CD* of concrete is then represented by Adaptive neuro-fuzzy inference system (*ANFIS*) models that we build and compare to the current models to see how successful they are. The *ANFIS* model is optimized in the present study using three reliable optimization methods: Bald Eagle Search (*BES*), Black Widow Optimization (*BWO*), and Artificial rabbit optimization (*ARO*). Numerous important variables are considered when estimating the *CD* of concrete, such as the water-to-binder ratio, the ratio of coarse aggregate to total aggregate, the ratio of silica fume to binder, the ratio of superplasticizer to binder, the curing process, the exposure duration, and the exposure condition. This work contributes to the area of civil engineering by developing a new model for concrete's *CD*. This instrument makes assessing the durability of concrete structures easier.

Research significance

Chloride diffusion is a primary cause of corrosion in reinforced concrete, especially in marine environments. This research provides a more reliable and long-term assessment of *CD* in concrete, offering critical insights for the design and maintenance of durable infrastructure such as bridges, ports, and seawalls. By understanding the real-time, long-term behavior of concrete in harsh environments, the study enhances the predictive

accuracy for structural durability, potentially reducing costly repairs and extending service life. Furthermore, many traditional *CD* models are based on short-term experimental data, which often fails to account for time-dependent changes in concrete properties. Tests like the *RCPT* have inherent drawbacks, such as high voltages leading to elevated temperatures and the inclusion of ions other than chloride. The research overcomes these limitations by utilizing long-term data and developing an *ANFIS* model that can more accurately represent *CD* over extended exposure periods in marine environments. Also, conducting long-term field tests on concrete structures is both time-consuming and expensive. By developing a robust *AI*-based predictive model, the study offers an alternative method to assess the durability of concrete structures efficiently, saving both time and resources while still maintaining high accuracy in predictions. Finally, the results from models can be directly applied by engineers to assess concrete performance in marine environments, informing better material selection and design practices.

Methods

Data description

A total of 216 data recordings were obtained from a long-term experimental field study on the durability of concrete under various exposure conditions. Seventy percent of the data (150 records) was utilized in the training section, whereas 30% (66 records) was utilized in the testing section [19–29]. The concrete samples were exposed to a range of atmospheric, splash, tidal, and submerged conditions for periods of 3 to 60 months; the training section exposure length was around 20 months, and the testing section exposure ranged from 3 to 84 months. The (a) water-to-binder ratio (W/B), (b) coarse aggregate to total aggregate (CAG/TAG), (c) silica fume-to-binder ratio (SF/B), (d) superplasticizer to binder ratio (SP/B), (e) curing mechanism (CM), (f) exposure time (ET), and (g) exposure condition (EC) are the seven essential parameters selected as input factors for creating the ensemble models. The concrete's *CD* is the sole outcome variable. Table 1 displays the data attributes of the original data and outcome variables. Every attribute are continuous variables, with the exception of EC . The formula for CM is $CM = M + 0.01N$, where M and N stand for the curing process and time (in days), accordingly. M is equal to 1 under the curing conditions of *air curing*, 2 under the curing conditions of *humid curing*, three under the curing conditions of *water curing*, and four under the curing conditions of *95% humidity curing*. The input and outcome distributions are shown in Fig. 1, along with an analysis of any potential outliers.

Bald Eagle Search (BES)

A novel heuristic method called the Bald Eagle Search method was created as a result of mother nature's inspiration [30]. The bald eagles intelligently seek their favorite foods, notably salmon fish. They are using a smart technique that involves choosing the best area where the target is present, exploring that area, and then diving in at the ideal moment to capture the victim. For an effective hunting technique, these bald eagles take advantage of the brisk wind and stormy weather.

The bald eagle's hunting strategy includes picking the ideal search area, exploring it, and diving down on its target at the perfect moment. The search area that an eagle often

Table 1 Statistical properties of attributes

Category	Statistics						
	Min.	Max.	St.d.	Var.	Avg.	Skew.	Kurt.
Inputs							
$\frac{W}{B}$							
Train	0.3	0.5	0.0681	0.00464	0.431	−0.397	−1.206
Test	0.3	0.5	0.0631	0.00398	0.428	−0.201	−1.1964
$\frac{CAG}{TAG}$							
Train	0.51	0.65	0.0316	0.001	0.554	1.8188	3.739
Test	0.51	0.65	0.0255	0.00065	0.552	2.43	7.887
$\frac{SF}{B}$							
Train	0	0.143	0.046	0.0021	0.0459	0.355	−1.283
Test	0	0.143	0.0481	0.0023	0.0445	0.41	−1.4043
$\frac{SP}{B}$							
Train	0	2.4	0.692	0.4791	0.609	1.306	0.899
Test	0	2.4	0.5486	0.301	0.4301	2.3038	5.9377
CM							
Train	2.07	3.28	0.444	0.1975	2.968	−1.382	0.2836
Test	2.07	3.28	0.4129	0.1705	3.0101	−1.648	1.352
ET							
Train	3	60	18.6314	347.129	19.81	0.8713	−0.533
Test	3	84	20.02	400.74	19.67	1.185	0.5944
Exposure type (count)							
EC	Tidal (T)		Splash (SP)	Atmosphere (A)	Submerged (SU)		
Train	84		32	21	11		
Test	30		22	9	5		
Target							
CD							
Train	0.21	27.55	4.552	20.722	3.667	2.373	6.8196
Test	0.21	21.79	4.2584	18.1346	3.4839	2.484	7.339

chooses is one that is closer to the last search area where it was successful in hunting. The eagle's prior motions influence every action it makes.

$$E_{new,i} = E_{best} + A' \times r(E_{mean} - E_i) \quad (1)$$

where E is the eagle's location, A' is a variable that controls how it changes its location, and it ranges from 1.5 to 2, and r accepts accidental values between 0 and 1. While E_{mean} takes into account whole prior search areas, E_{best} indicates the finest location in the past, which is now the most ideal. Bald eagles pick a search zone and then search it using helix patterns of different sizes.

$$E_{i,new} = E_i + y(i) \times (E_i - E_{i+1}) + x(i) \times (E_i - E_{mean}) \quad (2)$$

$$x(i) = \frac{xs(i)}{\max(|xs|)} \quad (3)$$

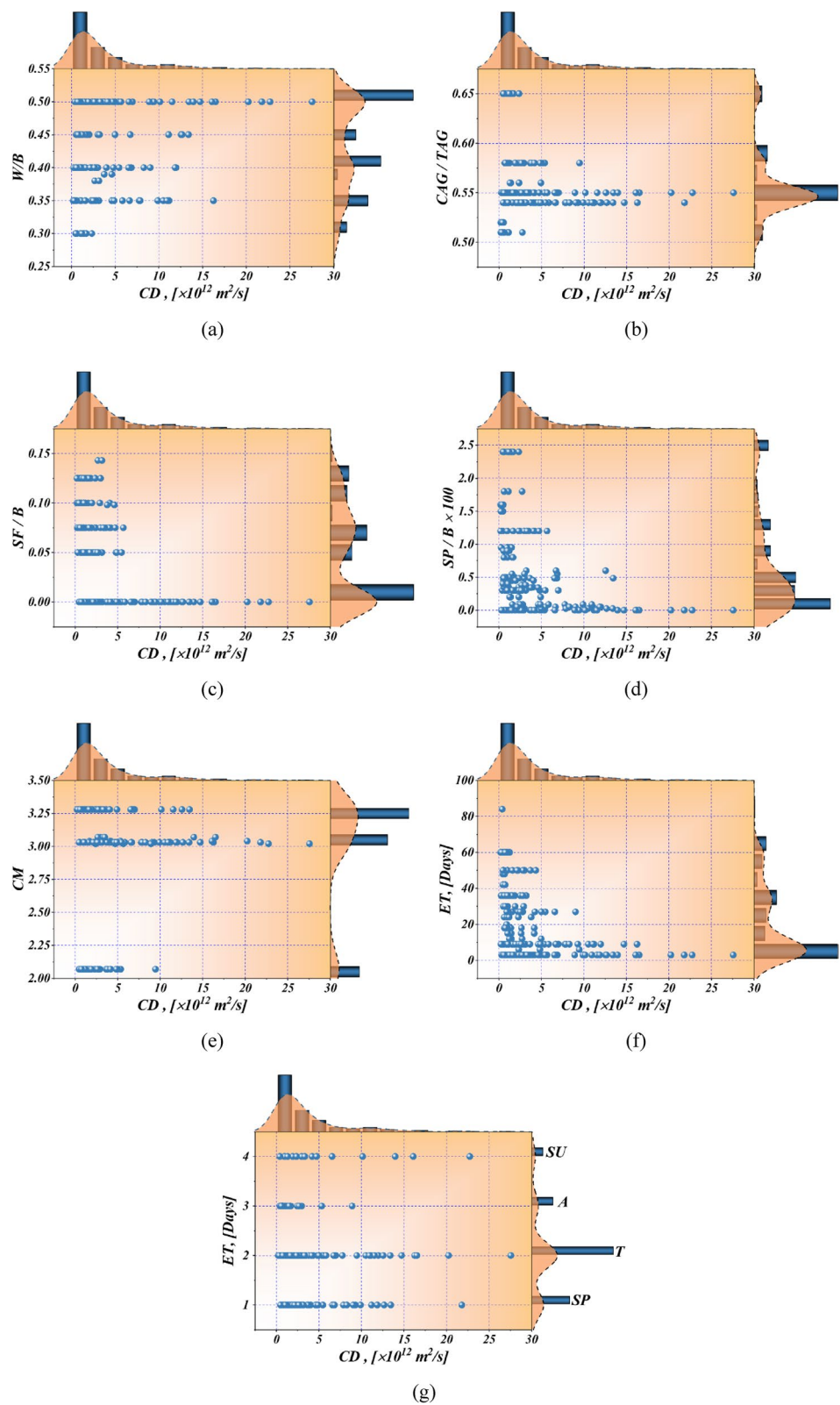


Fig. 1 The input attributes versus target

$$y(i) = \frac{ys(i)}{\max(|ys|)} \quad (4)$$

$$xs(i) = s(i) \times \sin\theta_i \quad (5)$$

$$ys(i) = s(i) \times \cos\theta_i \quad (6)$$

$$\theta_i = A'' \times \pi \times rd \quad (7)$$

$$s(i) = \theta_i + R \times rd \quad (8)$$

Here, R represents the number of search cycles that run from 0.5 to 2, rd indicates an accidental number, and A'' predicts the corner location and takes values between 5 and 10. Eagles dive on the target from the selected ideal location after exploring the area and determining the finest location to attack the prey. Mathematically, the diving phase can be expressed as follows:

$$E_{i,new} = rd \times E_{best} + xt(i) \times (E_i - b1 \times E_{mean}) + yt(i) \times (E_i - b2 \times E_{best}) \quad (9)$$

$$xt(i) = \frac{xs(i)}{\max(|xs|)} \quad (10)$$

$$yt(i) = \frac{ys(i)}{\max(|ys|)} \quad (11)$$

$$xs(i) = s(i) \times \sinh(\theta_i) \quad (12)$$

$$ys(i) = s(i) \times \cosh(\theta_i) \quad (13)$$

$$\theta_i = A'' \times \pi \times rd \quad (14)$$

$$s(i) = \theta_i \quad (15)$$

$b1$ and $b2$ accept values between 1 and 2. Figure 2 shows co-sequences for the three main steps of hunting by *BES*.

Black Widow Optimization (BWO)

Hayyolalam and Kazem invented the Black Widow method in 2020 to tackle challenging optimization challenges [31]. *BWO* belongs to the algorithm of evolution. The design of this method was inspired by the ways in which spiders reproduce, mutate, and choose among other things. Additionally, since this method offers quick convergence, it is discovered to perform superior to other evolutionary methods. This makes this method useful for solving a variety of challenging issues. The next part provides a detailed explanation of the black widow's natural behavior.

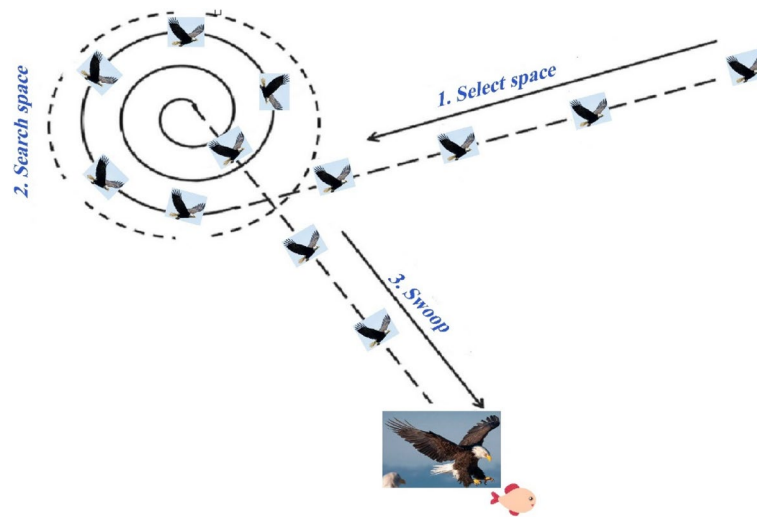


Fig. 2 Co-sequences for the three main steps of hunting by BES

As was mentioned in the preceding part, the natural behavior of black widows served as inspiration for the creation of *BWO*. This approach to tackling the optimization issues examined included many phases. The part that follows provides a thorough explanation of the steps involved in *BWO*.

Step 1: Initialization

The population is first established, including multiple widows of size M . For every widow in the population, there is a $1 \times M_{var}$ array. The issue's answer is shown in this array. The following is a mathematical representation of the provided array.

$$widow = (a_1, a_2, \dots, a_{M_{var}}) \quad (16)$$

The dimension associated with the optimization issue is denoted by $a_{M_{var}}$ in the formula provided earlier, along with the number of threshold values that must be collected using this technique, and the i_{th} solution is represented by a_i . Additionally, the fitness function of each widow is calculated to determine the widow's fitness. The following is a mathematical expression for the fitness function.

$$fitness = f(widow) \quad (17)$$

$$fitness = f(a_1, a_2, \dots, a_{M_{var}}) \quad (18)$$

Typically, the optimization procedure starts with the population being initialized in a size-based matrix ($M_{OP} \times M_{VAR}$). A pair of parents is then chosen from the population to continue the procedure of reproduction and mating.

Step 2: Procreation

This stage requires the introduction of an array named beta until the widow array containing an arbitrary parameter is present. To produce the novel offspring, a novel array must be generated. Usually, the parent pairings are unknown to one another, and they mate to

produce widows' novel generation. The alpha-created progeny of the next generation is provided in Formulas (9) and (10).

$$b_1 = \beta \times a_1 + (1 - \beta) \times a_2 \quad (19)$$

$$b_2 = \beta \times a_2 + (1 - \beta) \times a_1 \quad (20)$$

The novel generation of offspring is represented by b_1 and b_2 in the preceding formula, the parent widow is represented by a_1 and a_2 , and the enclosed array is represented by β . Ultimately, the recently created population is organized into an array and categorized according to its fitness value. After that, some of the finest widows are joined to a novel population according to cannibalism.

Step 3: Cannibalism

Three types of cannibalism have been identified in black widow spiders. In the first kind of cannibalism, known as sexual cannibalism, the female widow consumes her mate either during or after mating. Fitness values are used to identify the male and female widows. Sibling cannibalism is the second kind. The stronger spider progeny of this variety consumes the weaker one. According to their fitness value, the weaker and stronger offspring are categorized once again. The baby widow consumes their mother in the third kind of cannibalism.

Step 4: Mutation

The number of people chosen for mutation is determined by randomly exchanging two array items with one another. This process is known as mutation *mutepop*. Moreover, the mutation rate determines *mutepop*. The method is ended when the halting conditions are satisfied. Figure 3 presented the *BWO* schematic.

Artificial rabbit optimization (ARO)

ARO Is a novel metaheuristic optimizer inspired by the survival instinct of rabbits [32]. It employs two crucial survival tactics: hazardous hiding and detour foraging (Fig. 4). The mathematical formulas for the *ARO* approach are given in this section. When a rabbit is in foraging mode, it tries to eat the grasses and plants near other rabbits' burrows. This move protects the rabbit hole from attacks while also having the potential to fool hunters. In other words, rabbits avoid the easily available food in their immediate vicinity and seek food in distant areas. In the *ARO* approach, the number of rabbits is used as the swarm population. Each rabbit has an own feeding area with plants, grass, and b burrows. Every rabbit visits the burrows of the other rabbits in quest of food. Each rabbit has a tendency to adjust its current position in regard to the person who has been inadvertently selected during this mode and causes an annoyance. The following language might be used to describe this action:

$$\vec{x}_s(t+1) = \vec{y}_q(t) + p \cdot (\vec{y}_s(t) - \vec{y}_q(t)) + \text{round}(0.5 \cdot (0.05 + g_1)) \cdot \mu_1, s, q = 1, 2, \dots, \text{Rand}q \neq 1 \quad (21)$$

$$p = E \cdot e \quad (22)$$

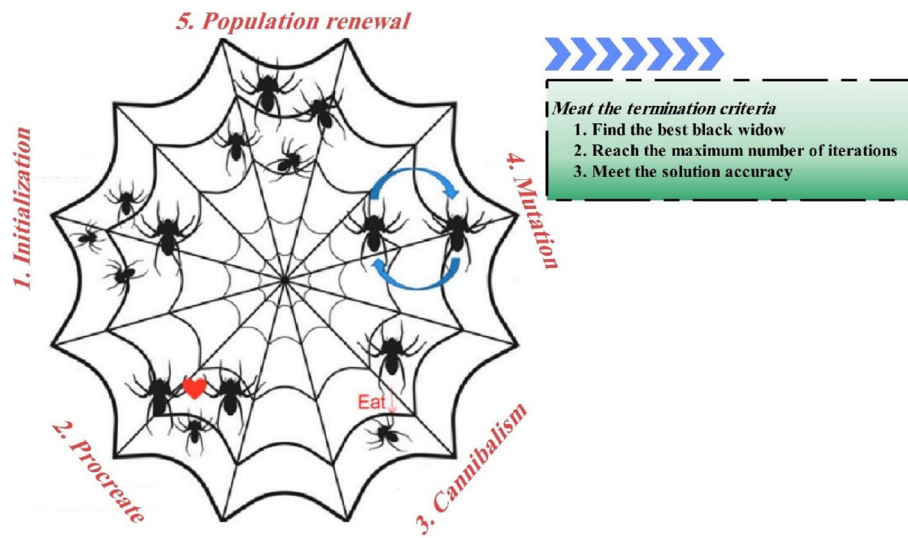


Fig. 3 Black Widow Optimization algorithm schematic

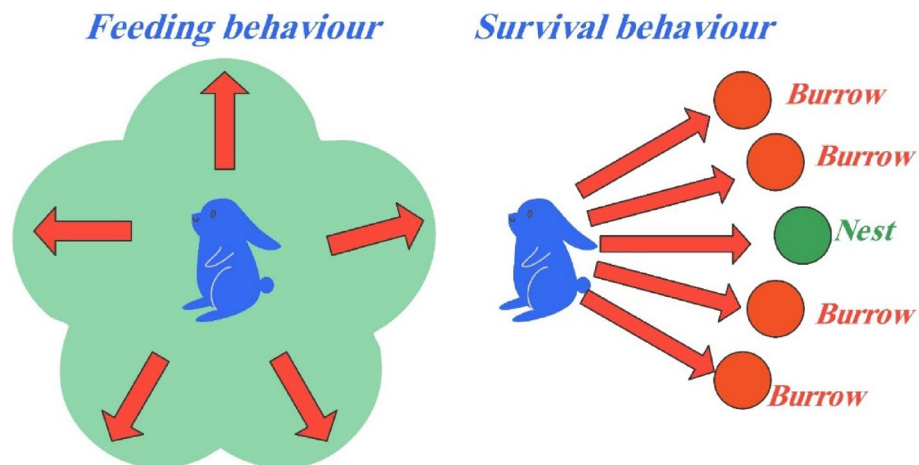


Fig. 4 The behavior of rabbits during optimization

$$E = \left(e - e^{\left(\frac{1-t}{T} \right)^2} \right) \bullet \sin(2\pi g_2) \quad (23)$$

$$f(w) = \begin{cases} 1 & \text{if } w == h(u) \\ 0 & \text{else} \end{cases} \quad k = 1, \dots, b \quad (24)$$

$$h = \text{randperm}(l) \quad (25)$$

$$n_1 \sim N(0,1) \quad (26)$$

In these equations, $\vec{x}_s(t+1)$ is the candidate's position of s_{th} individual at time $t+1$, $\vec{y}_s(t)$ is the s_{th} rabbit position at time t , R is for the rabbits' whole number, T is the iterations' whole number, round is round function to the amount with the closest

value, $randperm$ is the accidental permutation function, g is the accidental amount, E is the running interval throughout foraging, and n_1 is the normal diffusion function.

The change that is employed makes it possible for the ARO approach to effectively search completely and avoid local extrema. The running length function generates longer running intervals during the beginning rounds. The running lengths decrease with time, resulting in shorter lengths in the final iterations. Though the exploitation phase is associated with shorter running durations, the exploration phase is associated with longer running lengths. Throughout the search process, a number of people are selected using the vector f . The way the rabbits run is mimicked by the mathematical operator p . This foraging approach enhances the exploration and universal search capabilities of the ARO technique.

Rabbits randomly hide in the exploitation phase to avoid being discovered by hunters. The rabbit digs a few holes next to its holes. On each iteration, a rabbit digs b holes in the search region. To fool the hunters, it randomly selects a hole to hide in. This process is explained as follows:

$$B\vec{U}_{sq}(t) = \vec{y}_s(t) + H \bullet h \bullet \vec{y}_s(t), s = 1, 2, \dots, Randq = 1, 2, \dots, b \quad (27)$$

$$H = \frac{1 - t + T}{T} g_4 \quad (28)$$

$$n_2 \sim N(0,1) \quad (29)$$

$$\varphi(w) \begin{cases} 1 & \text{if } w == o \\ 0 & \text{else} \end{cases} \quad k = 1, \dots, b \quad (30)$$

In this context, H represents the concealed function, whereas b represents the created gaps in the search region. Over time, the area containing the created holes gradually decreases in size. The mathematical formulation of the rat's concealment approach is shown below:

$$\vec{x}_s(t+1) = \vec{y}_b(t) + p \bullet (g_4 \bullet BU_{sg}(t) - \vec{y}_b(t)), s = 1, 2, \dots, R \quad (31)$$

$$h_g(w) \begin{cases} 1 & \text{if } w == [g_5 \bullet b] \\ 0 & \text{else} \end{cases} \quad k = 1, \dots, b \quad (32)$$

$$BU_{sg}(t) = \vec{y}_s(t) + H \bullet h_g \bullet \vec{x}_s(t) \quad (33)$$

In the above equations, g_4 and g_5 are randomly selected numbers, and $BU_{sg}(t)$ indicates the specific hole that the rabbit has chosen to protect throughout the hiding process. Using Eq. (31), the i_{th} rabbit adjusts its position to reach a hole other than its own.

After the completion of the foraging and hiding procedures, the position of the i_{th} rabbit is sequentially adjusted in the following order:

$$\vec{y}_s(t+1) = \begin{cases} \vec{y}_s(t) & f(\vec{y}_s(t)) \leq f(\vec{x}_s(t+1)) \\ \vec{y}_s(t+1) & f(\vec{y}_s(t)) > f(\vec{x}_s(t+1)) \end{cases} \quad (34)$$

If the possible location is better than the current position, the location of the i_{th} rabbit is modified according to Eq. (34). In addition, the rabbits decrease their energy, which helps them transition from an exploratory behavior to an exploitative one. Rabbits do the foraging operation at the beginning of each iteration, whereas the concealment procedure is performed during the last iteration. The energy level of the rabbit influences its search behavior. The energy of the rabbit diminishes with time, and the potential energy sources might be as follows:

$$EF(t) = 4 \left(1 - \frac{t}{T} \right) \ln \left(\frac{1}{\sigma} \right) \quad (35)$$

In this section, an arbitrarily selected integer is denoted by σ . A rabbit has sufficient energy to engage in the process of foraging, as indicated by a significant quantity of the energy component. Due to the scarcity of energy resources, a rabbit is compelled to participate in the act of hiding itself, since it lacks the necessary energy to engage in the activity of foraging.

Adaptive neuro-fuzzy inference system (ANFIS)

The neuro-fuzzy network, known as *ANFIS*, was first developed in 1993 [33] and has subsequently been applied to a variety of scientific fields, including geosciences [34, 35]. The node variables determine the result of an adaptive network, and two different types of learning algorithms are used to control the variables throughout training in order to maximize the performance. In other terms, the *ANFIS* mechanism may be recognized as having two inputs (x and y) and one result (z). The outcome is calculated using five levels or stages that are given by if-then fuzzy rules according to the Sugeno model [36].

$$\text{Rule 1 : If } x \text{ is } A_1 \text{ and } y \text{ is } B_1, \text{ then } f_1 = p_1x + q_1y + r_1 \quad (36)$$

$$\text{Rule 2 : If } x \text{ is } A_2 \text{ and } y \text{ is } B_2, \text{ then } f_2 = p_2x + q_2y + r_2 \quad (36)$$

Here, A_1 , A_2 , B_1 , and B_2 are examples of fuzzy sets, whereas p_i , q_i , and r_i are examples of changeable variables. f_i denotes different results for the two rules.

Level 1: Each node I has a node function and is an adaptive node.

$$O_i^1 = \mu_{A_i}(x), i = 1, 2 \quad (38)$$

Or

$$O_i^1 = \mu_{B_{i-2}}(y), i = 3, 4 \quad (39)$$

Here, A_i (or B_{i-2}) is the linguistic label connected to the node, and x (or y) denotes the node i 's input. The membership function for A_i (or B_{i-2}), O_i^1 , or O_i^1 identifies the degree

to which the supplied x or y fulfills A_i (or B_{i-2}). According to [37], the A and B , which represent the membership functions, are often bell functions:

Typically, $A_i(x)$ is selected to be bell-shaped, with highest and lowest values of 1 and 0 correspondingly, as in the following:

$$\mu A_i(x) = \frac{1}{1 + \left| \frac{x-r_i}{p_i} \right|^{2q_i}} \quad (40)$$

In Eq. (11), p_i , q_i , and r_i are referred to as premise variables.

A suitable member for the node function at this level may be any continual and piece-wise identifiable function, such as a membership function with a triangular form. This level's variables are identified as premise variables.

Level 2: In level 2, each fixed node functions as a multiplier for the input signals and variables and demonstrates the firing intensity of the corresponding rule. The result displays as follows:

$$O_i^2 = w_i = \mu A_i(x) \times \mu B_i(y), i = 1, 2 \quad (41)$$

Level 3: Every node in level 3 relates to an adaptive node marked N , which calculates the ratio of the firing intensity of the i_{th} rule to the total firing intensity of the entire rules:

$$O_i^3 = \bar{w}_i = \frac{w_i}{w_1 + w_2}, i = 1, 2 \quad (42)$$

The results are typically identified as normalized firing intensities.

Level 4: Every node in level 4, referred to as an adaptive node, uses the following function to calculate the i_{th} rule's contribution to the total result:

$$O_i^4 = \bar{w}_i z_i = \bar{w}_i (a_i x + b_i y + c_i), i = 1, 2 \quad (43)$$

In Eq. (43), a_i , b_i , and c_i show the variable set, also known as resultant variables, and $\bar{w}_i$ is level 3's result.

Level 5: The single or fixed node computes the total result in level 5 by adding entire incoming/retrieved signals from levels before it as follows:

$$O_i^5 = \sum_i \bar{w}_i z_i = \frac{\sum_i w_i z_i}{\sum_i w_i} i = 1, 2 \quad (44)$$

Outline of coupled ANFISs

Data pretreatment, simulated training, and hyperparameter modification are crucial phases in the development of *ANFIS* models. The *ANFIS* optimization approach may assist in determining the optimal hyperparameters to enhance simulation performance. A description of the suggested procedure is provided below:

- a) Coding class characteristics and filling in missing values were part of cleaning and preparing the incoming dataset.
- b) The data collection process was divided into many training and testing groups in order to evaluate the process's effectiveness.
- c) A couple of hyperparameters (the total number of *MF* s and the number of fuzzy rules) were discovered to need tweaking.
- d) One target statistic that was thought to be useful as a productivity indicator was the *RMSE* function. It might be decreased or increased based on a certain set of hyperparameters.
- e) The best way to set up the hyperparameters was to use the *BES*, *BWO*, and *ARO*.
- f) *ANFBE*, *ANFBW*, and *ANFAR* models were constructed using the whole training dataset and the optimal hyperparameters that were found. Table 2 includes the original optimization method settings as well as the tuned hyperparameter values.
- g) Lastly, the testing samples were used to evaluate the precision and dependability.

Performance metrics

To assess the effectiveness of the *ANFAR*, *ANFBW*, and *ANFBE* models and facilitate comparison, eight efficiency factors were taken into consideration. These eight factors consist of the coefficient of determination (R^2), root-mean-square error (*RMSE*), relative absolute error (*RAE*), root relative square error (*RRSE*), mean absolute error (*MAE*), Theil inequality coefficient (*TIC*), agreement of forecasting results (*IA*), and normalized mean square (*NMSE*). Lower *RMSE*, *NRMSE*, *RAE*, *RRSE*, *TIC*, and *MAE* values

Table 2 The initial parameters of the optimization algorithms and tuned values of hyperparameters

Optimizers	Initial variables	Values	Hybrid	Properties	Values
<i>BES</i>	Iterations	150	<i>ANFBE</i>	Fuzzy rules	18
	Runs	10		<i>MF</i> s	12
	Populations	40		Epoch	50
	C_1 and C_2	2		Type of <i>MF</i>	<i>trimf</i>
	α	2		Shape of fuzzy	<i>Sugeno</i>
	R	1.5		Function	<i>Linear</i>
<i>BWO</i>	Iterations	150	<i>ANFBW</i>	Fuzzy rules	22
	Runs	10		<i>MF</i> s	19
	Populations	40		Epoch	50
	Procreate rate	0.3		Type of <i>MF</i>	<i>trimf</i>
	Mutation rate	0.5		Shape of fuzzy	<i>Sugeno</i>
				Function	<i>Linear</i>
<i>ARO</i>	Iterations	150	<i>ANFAR</i>	Fuzzy rules	28
	Runs	10		<i>MF</i> s	23
	Populations	40		Epoch	50
				Type of <i>MF</i>	<i>trimf</i>
				Shape of fuzzy	<i>Sugeno</i>
				Function	<i>Linear</i>

indicate a better degree accuracy. Moreover, better performance is indicated by greater R^2 and IA values.

$$R^2 = \left(\frac{\sum_{g=1}^G (n_g - \bar{n})(z_g - \bar{y})}{\sqrt{\left[\sum_{g=1}^G (n_g - \bar{n})^2 \right] \left[\sum_{g=1}^G (y_g - \bar{y})^2 \right]}} \right)^2 \quad (45)$$

$$NMSE = MSE/Var(n) \quad (46)$$

$$RMSE = \sqrt{\frac{1}{G} \sum_{g=1}^G (y_g - n_g)^2} \quad (47)$$

$$RAE = \frac{\sum_{g=1}^G |n_g - y_g|}{\sum_{g=1}^G |n_g - \bar{n}|} \quad (48)$$

$$RRSE = \sqrt{\frac{\sum_{g=1}^G (n_g - y_g)^2}{\sum_{g=1}^G (n_g - \bar{n})^2}} \quad (49)$$

$$MAE = \frac{1}{G} \sum_{g=1}^G |y_g - n_g| \quad (50)$$

$$TIC = \frac{\sqrt{\frac{1}{G} \sum_{g=1}^G (y_g - n_g)^2}}{(\sqrt{\frac{1}{G} \sum_{g=1}^G y_g^2} + \sqrt{\frac{1}{G} \sum_{g=1}^G n_g^2})} \quad (51)$$

$$IA = 1 - \frac{\sum_{g=1}^G (n_g - y_g)^2}{\sum_{g=1}^G (|n_g - \bar{n}| + |y_g - \bar{n}|)^2} \quad (52)$$

The measured D_c , mean of the measured D_c , predicted D_c , and the mean of the forecasted D_c are represented by the variables n_g , $\bar{n}$, y_g , and $\bar{y}$ in these formulas, accordingly. The dataset's row count is indicated by the G .

Results and discussions

Figure 5 shows the observed and predicted levels of concrete diffusion coefficient (CD) under different exposure conditions as a result of the testing and training steps of the recommended *ANFAR*, *ANFBW*, and *ANFBE* procedures; this study used the results of the *ANFAR*, *ANFBW*, and *ANFBE* techniques to determine the CD of concrete according to different exposure situations. Additionally, it is seen that the error rates in the figure follow a symmetrical distribution. The zero-error % line, indicating the percentage change in the CD focus, is perpendicular to the curve's center point. The metric R^2 , $RMSE$, $NMSE$, RAE , $RRSE$, MAE , TIC , and IA were utilized to evaluate the accuracy of

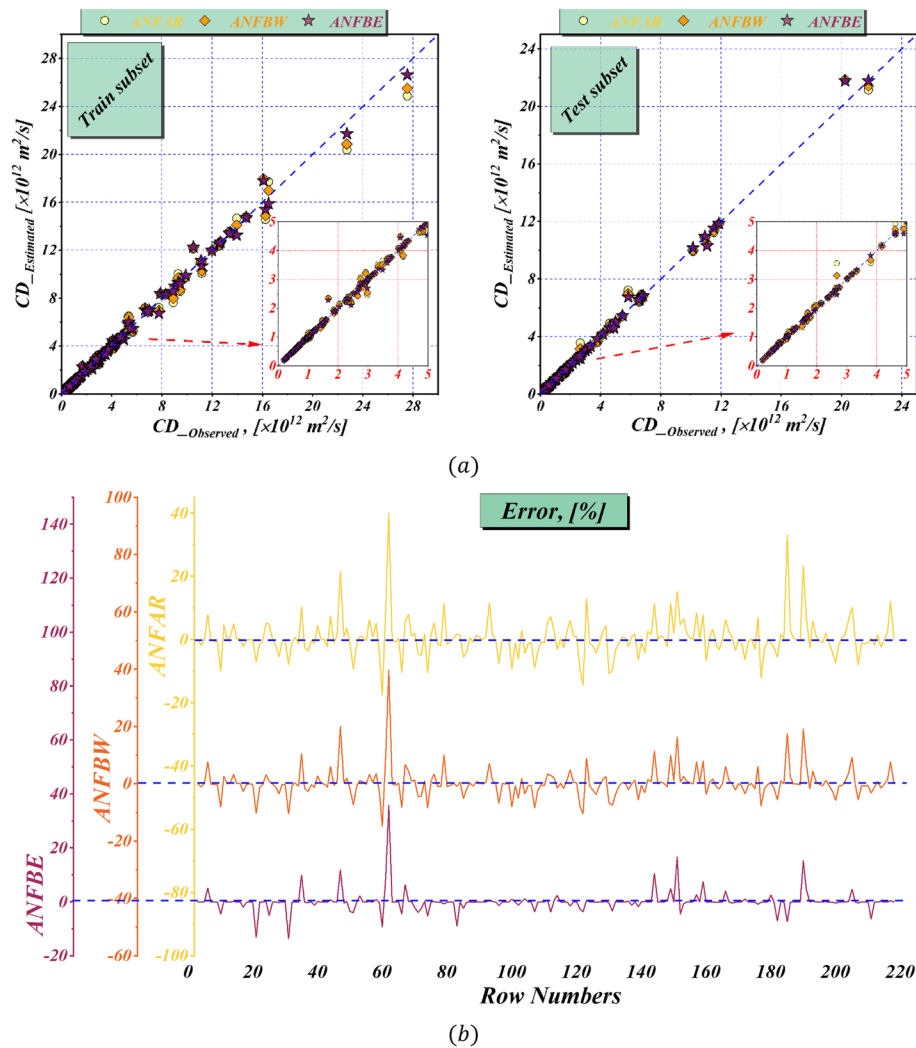


Fig. 5 The created systems' results

ANFAR, *ANFBW*, and *ANFBE* to forecast *CD* (see Table 3). In addition, this research conducted a comparison between the findings of the developed models and the most related research to evaluate the dependability and effectiveness of the models [16, 15].

The findings indicate that *ANFAR*, *ANFBW*, and *ANFBE* show considerable potential in accurately predicting the *CD* of concrete under different exposure conditions. *ANFBE* obtained an R^2 value of 0.9959 throughout the training section and 0.997 throughout the testing section. In particular, throughout the training section, *ANFBW* obtained an R^2 value of 0.9929, and throughout the testing section, it acquired a value of 0.996. The training section of *ANFAR* had an R^2 value of 0.99, while the testing section had a R^2 value of 0.995. The effectiveness of auxiliary techniques such as *NMSE*, *RAE*, *RRSE*, *MAE*, *TIC*, and *IA* must be thoroughly evaluated in order to achieve this goal. For the error-based metrics *RMSE*, *RRSE*, and *RAE*, the *ANFBE* model showed the smallest values. In comparison to the *ANFBW* and *ANFAR*, these figures were roughly 30% diminished. For example, according to the *RRSE* index, *ANFBE* had the lowest value during the training phase at 0.065, which was less than *ANFBW*'s score

Table 3 The created systems' results and literature

Created systems	Index							
	R^2	$RMSE$	RAE	$RRSE$	MAE	TIC	IA	$NMSE$
Train phase								
<i>ANFAR</i>	0.99	0.464	0.059	0.102	0.188	0.040	0.997	0.004
<i>ANFBW</i>	0.9929	0.390	0.048	0.086	0.150	0.034	0.998	0.003
<i>ANFBE</i>	0.9959	0.295	0.032	0.065	0.100	0.025	0.999	0.002
[16]	0.9033							
[15] (all data)	0.9495	1.033						
Test phase								
<i>ANFAR</i>	0.995	0.316	0.046	0.074	0.134	0.029	0.999	0.004
<i>ANFBW</i>	0.996	0.272	0.034	0.064	0.101	0.025	0.999	0.002
<i>ANFBE</i>	0.997	0.242	0.021	0.057	0.061	0.022	0.999	0.001
[16]	0.8815							

of 0.086 and *ANFAR*'s score of 0.102. Moreover, during the testing phase based on $RRSE$ values, *ANFBE* shows the smallest value at 0.057, which was less than *ANFBW* value at 0.064, and *ANFAR* value at 0.074.

Figure 5 displays the performance of the *ANFAR*, *ANFBW*, and *ANFBE* systems as a function of the error results among the forecasted and reported quantities. The evaluation encompasses both the learning and evaluation phases. A larger frequency of errors near the 0 line and narrower range plots indicate better accuracy and higher satisfying results. The *ANFBE* framework reveals a highly concentrated error range with limited upper and lower bounds, as the majority of samples cluster around the 0 line.

The literature is contrasted with the outcomes of the outperforming model (*ANFBE*) in the current found, as was previously indicated [15, 16]. For the sake of comparison, the R^2 and $RMSE$ metrics were used. In the training and testing portions, it is evident that the values of the R^2 have grown from 0.9033 [16] and 0.8815 [16] to 0.9959 and 0.997, respectively. This is a clear indication of the growth. The recent results of [15], obtained during the all-data phase, demonstrate a significant enhancement in performance via an increase in the R^2 value from 0.9495 to 0.9959 and a decrease in the $RMSE$ value from 1.033 to 0.0.295 (*ANFBE*).

Limitations and future recommendations

The amount and quality of the supplied data have a significant impact on the prediction models' accuracy and dependability. Inconsistent or missing data might negatively impact the model's performance. High-performance calculating resources may be needed for models that are complicated and computationally demanding due to the usage of advanced *AI* techniques like *ANFIS* and optimization approaches. To evaluate the predictive models' precision and dependability over time, their long-term efficacy in practical applications must be verified over a lengthy period of time.

For future modifications, some recommendations could be designed. To continually update and improve the prediction models according to actual field circumstances, real-time sensor data from embedded sensors in concrete structures can be used. The application of further sophisticated machine learning strategies to increase forecast resilience

and precision could be examined, such as deep learning, ensemble approaches, or hybrid models. The model's dependability and suitability for a variety of circumstances can be improved by regularly adding fresh study outcomes, field data, and empirical discoveries to the database from diverse sources. Finally, a risk assessment component that assesses tangible degradation implications based on expected CD values to prioritize maintenance and repair should be concluded.

Sensitivity analysis

Throughout this inquiry, sensitivity analysis is used to evaluate how input factors or variables impact performance. Model sensitivity analyses provide a systematic way to understand how input factors and variables affect the model's performance. The purpose of this is to improve decision-making, understanding, and model optimization. The effective model ($ANFBE$) was supplemented with a different set of input parameters for every one of the models that were constructed and presented in this study. To examine the effects of different inputs, R^2 , $RRSE$, and $NMSE$ metrics were created and compared with $ANFBE$, as illustrated in Table 4. The metric differences will increase in proportion to the degree to which missing components affect the result. The results indicate that the majority of the input factors ($\frac{W}{B}$, $\frac{CAG}{TAG}$, $\frac{SF}{B}$, $\frac{SP}{B}$, CM) have very little effect on the results when compared to the $ANFBE$. It is noteworthy that a notable increase in the $RRSE$ and $NMSE$, and a notable decrease in the R^2 value, occurs when variables related to ET and EC are removed from the input set. According to the elimination of EC , the R^2 value was decreased from 0.9959 to 0.8558; the $RRSE$ value was increased from 0.065 to 0.3798 in the learning phase. Also, on eliminating the ET , the $NMSE$ value was increased from 0.002 to 0.17 in the learning stage, and the R^2 value was decreased from 0.997 to 0.7761, the $RRSE$ value was increased from 0.057 to 0.4734, and the $NMSE$ value was increased from 0.001 to 0.1834 in the testing phase.

Conclusions

The scholars devised a method that combines an interconnected adaptive neuro-fuzzy inference system ($ANFIS$) to construct models capable of reliably predicting the diffusion chloride (CD) of concrete under various exposure conditions. The current research used the artificial rabbit optimization (ARO), Black Widow Optimization (BWO), and

Table 4 The results of various input scenarios using $ANFBE$

Input attributes	Train subset			Test subset		
	R^2	$RRSE$	$NMSE$	R^2	$RRSE$	$NMSE$
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SF}{B}, \frac{SP}{B}, CM, ET, EC$	0.9959	0.065	0.002	0.997	0.057	0.001
$\frac{CAG}{TAG}, \frac{SF}{B}, \frac{SP}{B}, CM, ET, EC$	0.9811	0.1384	0.0254	0.9771	0.1551	0.0308
$\frac{W}{B}, \frac{SF}{B}, \frac{SP}{B}, CM, ET, EC$	0.9959	0.0647	0.0017	0.9972	0.057	7.30E-04
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SP}{B}, CM, ET, EC$	0.9696	0.1749	0.0375	0.9608	0.1998	0.0975
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SF}{B}, CM, ET, EC$	0.9791	0.1473	0.0048	0.9764	0.1666	0.0101
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SF}{B}, \frac{SP}{B}, ET, EC$	>0.9508	0.2309	0.0137	0.9328	0.2941	0.0336
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SF}{B}, \frac{SP}{B}, CM, EC$	0.8622	0.3712	0.17	0.7761	0.4734	0.1834
$\frac{W}{B}, \frac{CAG}{TAG}, \frac{SF}{B}, \frac{SP}{B}, CM, ET$	0.8558	0.3798	0.1094	0.8513	0.3867	0.1624

Bald Eagle Search (*BES*) methods to identify important parameters in the *ANFIS* methodologies that may be enhanced. The research examines and compares the statistical measures used to assess the accuracy and reliability of each model. Furthermore, the unique sensitivity analysis approach is used to evaluate the effect of removing each variable on the aim. In addition, the current research assessed the reliability and effectiveness of the created models by comparing them with the most relevant research.

When *ET* and *EC* parameters are eliminated from the input set, *RRSE*, *NMSE*, and R^2 values significantly rise and drop. Eliminating *EC* led to a fall in R^2 from 0.9959 to 0.8558 and an increase in *RRSE* from 0.065 to 0.3798 during the learning phase. Eliminating *ET* led to increased *NMSE* values in learning (0.002 to 0.17), reduced R^2 values (0.997 to 0.7761), raised *RRSE* values (0.057 to 0.4734), and increased *NMSE* values (0.001 to 0.1834) in testing.

The results suggest that *ANFAR*, *ANFBW*, and *ANFBE* may reliably forecast concrete *CD* under varied exposure situations. *ANFBE* achieved R^2 values of 0.9959 during training and 0.997 during testing. During training, *ANFBW* achieved an R^2 value of 0.9929, while during testing, it reached 0.996. The training section of *ANFAR* has an R^2 value of 0.99, whereas the testing section had 0.995.

The *ANFBE* model has the lowest *RMSE*, *RRSE*, and *RAE* values. These values were 30% lower than *ANFBW* and *ANFAR*. For instance, *ANFBE* had the lowest *RRSE* index throughout training at 0.0065, behind *ANFBW*'s 0.086 and *ANFAR*'s 0.102. Additionally, throughout testing, *ANFBE* had the lowest *RRSE* score at 0.057, followed by *ANFBW* at 0.064 and *ANFAR* at 0.074.

For comparison, R^2 and *RMSE* measurements were utilized. Both training and testing phases show increased R^2 values from 0.9033 (literature) to 0.9959 and 0.997, respectively. This shows growth. The all-data phase results of the literature show a considerable improvement in performance, with a gain in R^2 from 0.9495 to 0.9959 and a drop in *RMSE* from 1.033 to 0.0.295 (*ANFBE*).

This study gives a more accurate and long-term measurement of *CD* in concrete, which is important for designing and maintaining bridges, ports, and seawalls. The work improves structural durability prediction by studying concrete's real-time, long-term behavior in severe conditions, saving expensive repairs, and increasing service life. Long-term data and an *ANFIS* model that better represents *CD* during marine exposure periods solve these shortcomings. Long-term field testing on concrete structures is costly and time-consuming. The research developed a strong *AI*-based predictive model to effectively analyze concrete structure durability, saving time and money while retaining high accuracy. Finally, engineers may use model findings to evaluate concrete performance in maritime conditions, improving material selection and design.

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Authors' contributions

The author contributed to the study's conception and design. Data collection, simulation, and analysis were performed by "SS." Also, the first draft of the manuscript was written by SS who commented on previous versions of the manuscript. The author read and approved the final manuscript.

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Data availability

Data can be shared upon request.

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The author declares no competing interests.

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